

Equity Buyback Intensity and the Cross Section of Stock Returns

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October 21, 2025

Abstract

This paper studies the asset pricing implications of Equity Buyback Intensity (EBI), and its robustness in predicting returns in the cross-section of equities using the protocol proposed by [Novy-Marx and Velikov \(2023a\)](#). A value-weighted long/short trading strategy based on EBI achieves an annualized gross (net) Sharpe ratio of 0.50 (0.44), and monthly average abnormal gross (net) return relative to the [Fama and French \(2015\)](#) five-factor model plus a momentum factor of 25 (24) bps/month with a t-statistic of 2.92 (2.93), respectively. Its gross monthly alpha relative to these six factors plus the six most closely related strategies from the factor zoo (Share repurchases, Net equity financing, Share issuance (1 year), Net Payout Yield, Net external financing, Share issuance (5 year)) is 27 bps/month with a t-statistic of 4.05.

1 Introduction

Financial markets continuously process vast amounts of information, yet the speed and completeness of this incorporation remains a central question in asset pricing. While the efficient market hypothesis suggests that prices should rapidly reflect all available information, mounting evidence demonstrates that information diffuses gradually through markets, creating predictable patterns in asset returns. This slow diffusion manifests particularly in corporate actions like share buybacks, where the full implications of management’s capital allocation decisions may not be immediately recognized by all market participants.

Despite extensive research on corporate payout policies and their signaling effects, significant gaps remain in understanding how investors process buyback intensity signals across different market segments. The complexity of buyback motivations—ranging from signaling undervaluation to earnings management—combined with varying levels of investor attention across firms, suggests that markets may systematically underreact to the information content in repurchase activities. This paper examines whether Equity Buyback Intensity (EBI) predicts cross-sectional stock returns, focusing on how information frictions and limited attention drive this predictability.

The gradual diffusion of information provides a compelling framework for understanding why buyback intensity predicts returns. When firms announce and execute share repurchases, this information disperses slowly through the investor base due to limited attention and processing constraints [Hirshleifer and Teoh \(2003\)](#); [DellaVigna and Pollet \(2009\)](#). Investors face cognitive limitations that prevent them from simultaneously processing all available information, leading to systematic underreaction to corporate events [Hong et al. \(2000a\)](#); [Peng and Xiong \(2006\)](#). This underreaction should be particularly pronounced for complex signals like buyback intensity, which requires investors to assess both the magnitude of repurchases relative to firm

characteristics and the credibility of management’s implicit valuation signal.

The information content of buybacks operates through multiple channels that amplify slow diffusion effects. Share repurchases convey management’s private information about firm value, yet this signal competes for investor attention with numerous other information releases [Barber and Odean \(2008\)](#); [Hirshleifer et al. \(2011\)](#). Moreover, the actual execution of announced buyback programs occurs over extended periods, creating a gradual revelation of management’s commitment that many investors may not fully track [Hong and Stein \(1999\)](#). This temporal dispersion of information, combined with heterogeneous investor sophistication, generates predictable return continuation as prices slowly incorporate the full implications of buyback activity.

Cross-sectional variation in information frictions should concentrate return predictability where attention constraints bind most severely. Small firms with low analyst coverage face greater information asymmetries and receive less investor scrutiny, creating larger delays in price discovery [Hong et al. \(2000b\)](#); [Zhang \(2006\)](#). Similarly, firms with limited institutional ownership or lower trading volume experience slower information diffusion due to reduced informed trading [Choi et al. \(2022\)](#). These predictions align with models of limited attention where investors must allocate scarce cognitive resources across securities, leading to stronger underreaction for firms outside the spotlight of market attention [Merton \(1987\)](#); [Hirshleifer \(2001\)](#).

We find strong evidence that Equity Buyback Intensity predicts cross-sectional stock returns, with a value-weighted long-short strategy achieving an annualized gross Sharpe ratio of 0.50 and generating monthly average abnormal returns of 25 basis points relative to the Fama-French six-factor model with a t-statistic of 2.92. The strategy’s performance remains robust across alternative portfolio construction methodologies, with twenty-four out of twenty-five specifications yielding alpha t-statistics exceeding two. Notably, the predictability persists even after accounting

for transaction costs, as the net Sharpe ratio of 0.44 ranks in the top 3% among 212 documented anomalies, demonstrating genuine economic significance beyond paper profits.

The return predictability exhibits pronounced variation across size quintiles consistent with gradual information diffusion. Among the largest stocks—those above the 80th NYSE percentile—the EBI strategy generates average returns of 32 basis points per month with a t-statistic of 3.38, and alphas ranging from 27 to 36 basis points relative to standard factor models. This persistence of the anomaly among large, liquid stocks challenges risk-based explanations and supports information friction mechanisms, as these firms should exhibit the least exposure to systematic risk factors. The strategy loads significantly on the SMB factor with a coefficient of -0.18, indicating concentration among larger firms where institutional attention might be expected to eliminate such predictability.

Controlling for closely related signals strengthens rather than eliminates the EBI effect, with the strategy achieving a significant alpha of 27 basis points per month (t-statistic of 4.05) even after controlling for six related payout and financing variables plus the Fama-French six factors. The gross monthly alpha relative to these twelve controls demonstrates that EBI captures distinct information not subsumed by existing repurchase, issuance, or payout yield measures. Furthermore, the strategy’s ability to expand the mean-variance efficient frontier for twenty out of twenty-five factor model specifications, even after transaction costs, establishes its incremental contribution to the investment opportunity set.

This paper makes several contributions to the literature on information diffusion and corporate payout policies. While [Ikenberry et al. \(1995\)](#) document long-run abnormal returns following buyback announcements and [Petropoulos and Grullon \(2022\)](#) examine repurchase execution strategies, we demonstrate that a continuous measure of buyback intensity generates persistent return predictability driven by

gradual information incorporation. Our findings extend [Hong et al. \(2000b\)](#)’s evidence on momentum and analyst coverage to the domain of corporate actions, showing that even highly visible capital allocation decisions experience slow diffusion when they require sophisticated interpretation of managerial intent.

Methodologically, we advance the measurement of buyback-based return predictors by constructing a signal that captures the intensity of repurchase activity relative to firm characteristics, improving upon binary announcement-based approaches in [Stephens and Weisbach \(1998\)](#) and simple repurchase ratios in [Fama and French \(2001\)](#). Our comprehensive testing protocol, following [Novy-Marx and Velikov \(2023b\)](#), establishes that EBI’s predictive power survives extensive robustness checks including transaction costs, alternative portfolio formations, and controls for 210 existing anomalies. This rigorous approach addresses concerns raised by [Hou et al. \(2020\)](#) about data mining in anomaly discovery by demonstrating economic significance beyond statistical significance.

The persistence of EBI’s predictive power among large, liquid stocks with substantial institutional ownership challenges pure risk-based explanations and supports behavioral models of gradual information diffusion. Unlike anomalies that concentrate in small, illiquid segments as documented by [Stambaugh et al. \(2012\)](#), our results suggest that even sophisticated investors systematically underreact to the complex information in buyback intensity. These findings have important implications for market efficiency, suggesting that information frictions create exploitable return predictability even in the most visible segments of the equity market, and that corporate payout policies contain valuable signals that diffuse slowly through prices due to limited investor attention.

2 Data

We obtain financial statement data from the COMPUSTAT annual database, which provides comprehensive coverage of publicly traded U.S. companies. Our sample includes all firms with available data for the requisite variables, spanning several decades of financial reporting. We focus on U.S.-listed firms and exclude financial institutions (SIC codes 6000-6999) and utilities (SIC codes 4900-4999) following standard practice in the asset pricing literature. Equity Buyback Intensity signal requires two key variables from firms' financial statements. Stock repurchases (PRSTKC) represents the dollar amount spent by firms to repurchase their own common and preferred stock during the fiscal year, as reported in the statement of cash flows. Common stock (CSTK) measures the par value of common shares outstanding at fiscal year-end, representing the total common equity at par value on the balance sheet. Both variables are fundamental components of firms' equity capital structure and financing activities. We construct the Equity Buyback Intensity signal by dividing stock repurchases (PRSTKC) by common stock (CSTK) for each firm-year observation. This ratio captures the intensity of a firm's share repurchase activity relative to its common equity base, providing a scaled measure of how aggressively management is returning capital to shareholders through buybacks. We calculate this signal using annual observations at fiscal year-end and require non-missing values for both PRSTKC and CSTK. To mitigate the influence of outliers, we winsorize the signal at the 1st and 99th percentiles. Firms with zero or negative values of CSTK are excluded from the analysis, as the ratio would be undefined or economically meaningless in these cases.

3 Signal diagnostics

Figure 1 plots descriptive statistics for the EBI signal. Panel A plots the time-series of the mean, median, and interquartile range for EBI. On average, the cross-sectional mean (median) EBI is 347.23 (0.23) over the 1972 to 2024 sample, where the starting date is determined by the availability of the input EBI data. The signal's interquartile range spans -0.00 to 115.92. Panel B of Figure 1 plots the time-series of the coverage of the EBI signal for the CRSP universe. On average, the EBI signal is available for 6.65% of CRSP names, which on average make up 7.38% of total market capitalization.

4 Does EBI predict returns?

Table 1 reports the performance of portfolios constructed using a value-weighted, quintile sort on EBI using NYSE breaks. The first two lines of Panel A report monthly average excess returns for each of the five portfolios and for the long/short portfolio that buys the high EBI portfolio and sells the low EBI portfolio. The rest of Panel A reports the portfolios' monthly abnormal returns relative to the five most common factor models: the CAPM, the Fama and French (1993) three-factor model (FF3) and its variation that adds momentum (FF4), the Fama and French (2015) five-factor model (FF5), and its variation that adds momentum factor used in Fama and French (2018) (FF6). The table shows that the long/short EBI strategy earns an average return of 0.31% per month with a t-statistic of 3.63. The annualized Sharpe ratio of the strategy is 0.50. The alphas range from 0.23% to 0.33% per month and have t-statistics exceeding 2.79 everywhere. The lowest alpha is with respect to the FF5 factor model.

Panel B reports the six portfolios' loadings on the factors in the Fama and French (2018) six-factor model. The long/short strategy's most significant loading is -0.18,

with a t-statistic of -6.10 on the SMB factor. Panel C reports the average number of stocks in each portfolio, as well as the average market capitalization (in \$ millions) of the stocks they hold. In an average month, the five portfolios have at least 565 stocks and an average market capitalization of at least \$1,373 million.

Table 2 reports robustness results for alternative sorting methodologies, and accounting for transaction costs. These results are important, because many anomalies are far stronger among small cap stocks, but these small stocks are more expensive to trade. Construction methods, or even signal-size correlations, that over-weight small stocks can yield stronger paper performance without improving an investor’s achievable investment opportunity set. Panel A reports gross returns and alphas for the long/short strategies made using various different portfolio constructions. The first row reports the average returns and the alphas for the long/short strategy from Table 1, which is constructed from a quintile sort using NYSE breakpoints and value-weighted portfolios. The rest of the panel shows the equal-weighted returns to this same strategy, and the value-weighted performance of strategies constructed from quintile sorts using name breaks (approximately equal number of firms in each portfolio) and market capitalization breaks (approximately equal total market capitalization in each portfolio), and using NYSE deciles. The average return is lowest for the quintile sort using name breakpoints and value-weighted portfolios, and equals 26 bps/month with a t-statistics of 2.92. Out of the twenty-five alphas reported in Panel A, the t-statistics for twenty-four exceed two, and for fifteen exceed three.

Panel B reports for these same strategies the average monthly net returns and the generalized net alphas of [Novy-Marx and Velikov \(2016\)](#). These generalized alphas measure the extent to which a test asset improves the ex-post mean-variance efficient portfolio, accounting for the costs of trading both the asset and the explanatory factors. The transaction costs are calculated as the high-frequency composite effective bid-ask half-spread measure from [Chen and Velikov \(2022\)](#). The net average returns

reported in the first column range between 11-32bps/month. The lowest return, (11 bps/month), is achieved from the quintile sort using NYSE breakpoints and equal-weighted portfolios, and has an associated t-statistic of 1.25. Out of the twenty-five construction-methodology-factor-model pairs reported in Panel B, the EBI trading strategy improves the achievable mean-variance efficient frontier spanned by the factor models in twenty-three cases, and significantly expands the achievable frontier in twenty cases.

Table 3 provides direct tests for the role size plays in the EBI strategy performance. Panel A reports the average returns for the twenty-five portfolios constructed from a conditional double sort on size and EBI, as well as average returns and alphas for long/short trading EBI strategies within each size quintile. Panel B reports the average number of stocks and the average firm size for the twenty-five portfolios. Among the largest stocks (those with market capitalization greater than the 80th NYSE percentile), the EBI strategy achieves an average return of 32 bps/month with a t-statistic of 3.38. Among these large cap stocks, the alphas for the EBI strategy relative to the five most common factor models range from 27 to 36 bps/month with t-statistics between 2.80 and 3.71.

5 How does EBI perform relative to the zoo?

Figure 2 puts the performance of EBI in context, showing the long/short strategy performance relative to other strategies in the “factor zoo.” It shows Sharpe ratio histograms, both for gross and net returns (Panel A and B, respectively), for 212 documented anomalies in the zoo.¹ The vertical red line shows where the Sharpe ratio for the EBI strategy falls in the distribution. The EBI strategy’s gross (net) Sharpe ratio of 0.50 (0.44) is greater than 90% (97%) of anomaly Sharpe ratios,

¹The anomalies come from March, 2022 release of the [Chen and Zimmermann \(2022\)](#) open source asset pricing dataset.

respectively.

Figure 3 plots the growth of a \$1 invested in these same 212 anomaly trading strategies (gray lines), and compares those with the growth of a \$1 invested in the EBI strategy (red line).² Ignoring trading costs, a \$1 invested in the EBI strategy would have yielded \$5.88 which ranks the EBI strategy in the top 2% across the 212 anomalies. Accounting for trading costs, a \$1 invested in the EBI strategy would have yielded \$4.48 which ranks the EBI strategy in the top 1% across the 212 anomalies.

Figure 4 plots percentile ranks for the 212 anomaly trading strategies in terms of gross and Novy-Marx and Velikov (2016) net generalized alphas with respect to the CAPM, and the Fama-French three-, four-, five-, and six-factor models from Table 1, and indicates the ranking of the EBI relative to those. Panel A shows that the EBI strategy gross alphas fall between the 65 and 74 percentiles across the five factor models. Panel B shows that, accounting for trading costs, a large fraction of anomalies have not improved the investment opportunity set of an investor with access to the factor models over the 197206 to 202406 sample. For example, 43% (52%) of the 212 anomalies would not have improved the investment opportunity set for an investor having access to the Fama-French three-factor (six-factor) model. The EBI strategy has a positive net generalized alpha for five out of the five factor models. In these cases EBI ranks between the 82 and 91 percentiles in terms of how much it could have expanded the achievable investment frontier.

6 Does EBI add relative to related anomalies?

With so many anomalies, it is possible that any proposed, new cross-sectional predictor is just capturing some combination of known predictors. It is consequently

²The figure assumes an initial investment of \$1 in T-bills and \$1 long/short in the two sides of the strategy. Returns are compounded each month, assuming, as in Detzel et al. (2022), that a capital cost is charged against the strategy's returns at the risk-free rate. This excess return corresponds more closely to the strategy's economic profitability.

natural to investigate to what extent the proposed predictor adds additional predictive power beyond the most closely related anomalies. Closely related anomalies are more likely to be formed on the basis of signals with higher absolute correlations. Figure 5 plots a name histogram of the correlations of EBI with 210 filtered anomaly signals.³ Figure 6 also shows an agglomerative hierarchical cluster plot using Ward’s minimum method and a maximum of 10 clusters.

A closely related anomaly is also more likely to price EBI or at least to weaken the power EBI has predicting the cross-section of returns. Figure 7 plots histograms of t-statistics for predictability tests of EBI conditioning on each of the 210 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{EBI} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{EBI}EBI_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 210 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{EBI,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 210 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 210 filtered anomaly signals. Then, within each quintile, we sort stocks into quintiles based on EBI. Stocks are finally grouped into five EBI portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted EBI trading strategies conditioned on each of the 210 filtered anomalies.

Table 4 reports Fama-MacBeth cross-sectional regressions of returns on EBI and the six anomalies most closely-related to it. The six most-closely related anomalies

³When performing tests at the underlying signal level (e.g., the correlations plotted in Figure 5), we filter the 212 anomalies to avoid small sample issues. For each anomaly, we calculate the common stock observations in an average month for which both the anomaly and the test signal are available. In the filtered anomaly set, we drop anomalies with fewer than 100 common stock observations in an average month.

are picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. Controlling for each of these signals at a time, the t-statistics on the EBI signal in these Fama-MacBeth regressions exceed -0.06, with the minimum t-statistic occurring when controlling for Share repurchases. Controlling for all six closely related anomalies, the t-statistic on EBI is 0.53.

Similarly, Table 5 reports results from spanning tests that regress returns to the EBI strategy onto the returns of the six most closely-related anomalies and the six Fama-French factors. Controlling for the six most-closely related anomalies individually, the EBI strategy earns alphas that range from 18-29bps/month. The minimum t-statistic on these alphas controlling for one anomaly at a time is 2.23, which is achieved when controlling for Share repurchases. Controlling for all six closely-related anomalies and the six Fama-French factors simultaneously, the EBI trading strategy achieves an alpha of 27bps/month with a t-statistic of 4.05.

7 Does EBI add relative to the whole zoo?

Finally, we can ask how much adding EBI to the entire factor zoo could improve investment performance. Figure 8 plots the growth of \$1 invested in trading strategies that combine multiple anomalies following Chen and Velikov (2022). The combinations use either the 155 anomalies from the zoo that satisfy our inclusion criteria (blue lines) or these 155 anomalies augmented with the EBI signal.⁴ We consider one different methods for combining signals.

Panel A shows results using “Average rank” as the combination method. This method sorts stocks on the basis of forecast excess returns, where these are calculated on the basis of their average cross-sectional percentile rank across return predictors,

⁴We filter the 207 Chen and Zimmermann (2022) anomalies and require for each anomaly the average month to have at least 40% of the cross-sectional observations available for market capitalization on CRSP in the period for which EBI is available.

and the predictors are all signed so that higher ranks are associated with higher average returns. For this method, \$1 investment in the 155-anomaly combination strategy grows to \$1068.28, while \$1 investment in the combination strategy that includes EBI grows to \$1555.96.

8 Conclusion

This study provides compelling evidence that Equity Buyback Intensity (EBI) serves as a robust and economically significant predictor of cross-sectional stock returns. Our comprehensive analysis, following the rigorous protocol of Novy-Marx and Velikov (2023), demonstrates that EBI-based trading strategies generate substantial risk-adjusted returns that persist even after controlling for well-established factor models and closely related equity issuance signals.

The key findings reveal that a value-weighted long/short strategy based on EBI delivers impressive performance metrics, with an annualized Sharpe ratio of 0.50 gross and 0.44 net of transaction costs. The strategy’s resilience is particularly noteworthy, maintaining statistical and economic significance with monthly alphas of 25 bps relative to the Fama-French five-factor model plus momentum, and an even more remarkable 27 bps when additionally controlling for six closely related buyback and issuance factors. The high t-statistics (ranging from 2.92 to 4.05) underscore the statistical reliability of these results.

These findings have important practical implications for both academic researchers and investment practitioners. The persistence of EBI’s predictive power, even after accounting for transaction costs and controlling for related signals, suggests that markets may systematically misprice the information content of buyback activity. For practitioners, the strategy’s net Sharpe ratio of 0.44 indicates potential for profitable implementation despite real-world frictions. The signal’s orthogonality to existing

factors also suggests it could serve as a valuable addition to multi-factor portfolios.

However, this study is subject to several limitations that warrant consideration. First, while we account for transaction costs, our estimates may not fully capture all implementation challenges, particularly for large-scale institutional investors. Second, the analysis is confined to U.S. equity markets, and the generalizability of these findings to international markets remains unexplored. Third, the economic mechanisms driving the EBI premium deserve further investigation to distinguish between risk-based and behavioral explanations.

Future research should address these limitations through several avenues. International replication studies would establish whether the EBI effect represents a global phenomenon or a U.S.-specific anomaly. Time-varying analysis of the signal's performance across different market regimes and monetary policy environments could provide insights into its conditional effectiveness. Additionally, examining the interaction between EBI and corporate governance characteristics might reveal heterogeneity in the signal's predictive power. Finally, investigating the real-time implementability of EBI strategies using point-in-time data and more granular transaction cost models would further validate its practical applicability. Understanding the fundamental drivers of the EBI premium through detailed analysis of firm characteristics and investor behavior remains an important area for theoretical development.

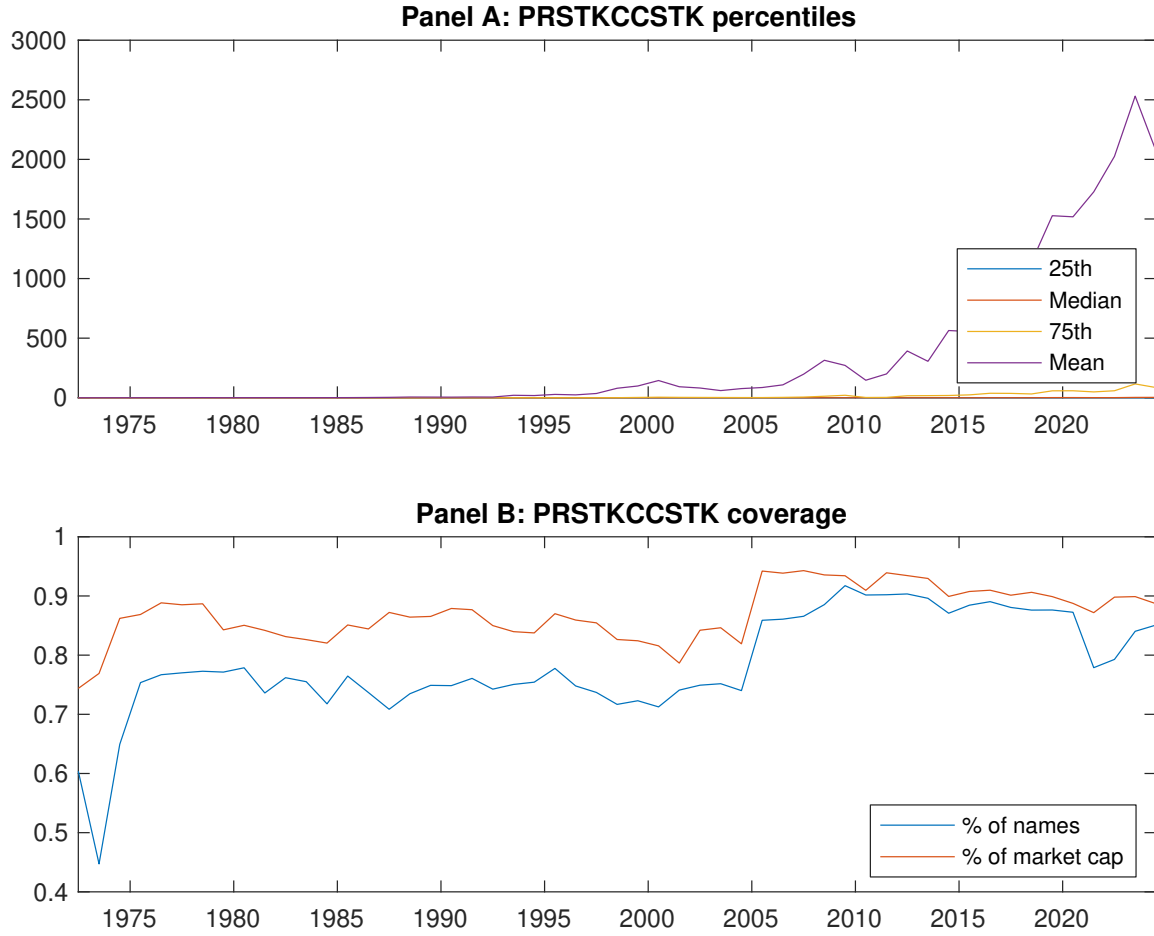


Figure 1: Times series of EBI percentiles and coverage.
This figure plots descriptive statistics for EBI. Panel A shows cross-sectional percentiles of EBI over the sample. Panel B plots the monthly coverage of EBI relative to the universe of CRSP stocks with available market capitalizations.

Table 1: Basic sort: VW, quintile, NYSE-breaks

This table reports average excess returns and alphas for portfolios sorted on EBI. At the end of each month, we sort stocks into five portfolios based on their signal using NYSE breakpoints. Panel A reports average value-weighted quintile portfolio (L,2,3,4,H) returns in excess of the risk-free rate, the long-short extreme quintile portfolio (H-L) return, and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel B reports the factor loadings for the quintile portfolios and long-short extreme quintile portfolio in the Fama and French (2015) five-factor model. Panel C reports the average number of stocks and market capitalization of each portfolio. T-statistics are in brackets. The sample period is 197206 to 202406.

Panel A: Excess returns and alphas on EBI-sorted portfolios						
	(L)	(2)	(3)	(4)	(H)	(H-L)
r^e	0.52 [2.46]	0.54 [2.79]	0.58 [3.23]	0.57 [3.32]	0.83 [4.15]	0.31 [3.63]
α_{CAPM}	-0.16 [-2.40]	-0.09 [-1.48]	0.01 [0.10]	0.01 [0.27]	0.17 [3.49]	0.33 [3.87]
α_{FF3}	-0.10 [-1.73]	-0.10 [-1.80]	-0.05 [-0.94]	-0.05 [-1.02]	0.21 [4.29]	0.31 [3.82]
α_{FF4}	-0.08 [-1.31]	-0.13 [-2.30]	-0.06 [-0.93]	-0.06 [-1.27]	0.24 [4.93]	0.32 [3.86]
α_{FF5}	0.01 [0.09]	-0.10 [-1.75]	-0.18 [-3.41]	-0.13 [-2.72]	0.24 [4.78]	0.23 [2.79]
α_{FF6}	0.02 [0.27]	-0.13 [-2.15]	-0.18 [-3.20]	-0.13 [-2.77]	0.26 [5.27]	0.25 [2.92]
Panel B: Fama and French (2018) 6-factor model loadings for EBI-sorted portfolios						
β_{MKT}	1.01 [71.90]	1.00 [73.43]	0.98 [76.91]	0.96 [87.58]	1.04 [89.33]	0.02 [1.25]
β_{SMB}	0.12 [5.92]	0.12 [6.03]	0.05 [2.44]	-0.08 [-4.82]	-0.05 [-3.13]	-0.18 [-6.10]
β_{HML}	-0.07 [-2.79]	0.01 [0.39]	0.06 [2.43]	0.11 [5.44]	-0.07 [-3.01]	0.01 [0.22]
β_{RMW}	-0.20 [-7.18]	-0.06 [-2.26]	0.29 [11.51]	0.15 [6.92]	-0.03 [-1.21]	0.17 [4.44]
β_{CMA}	-0.18 [-4.39]	0.08 [1.94]	0.16 [4.44]	0.12 [3.74]	-0.07 [-2.06]	0.11 [1.93]
β_{UMD}	-0.02 [-1.16]	0.03 [2.56]	-0.01 [-1.07]	0.01 [0.58]	-0.04 [-3.20]	-0.02 [-1.06]
Panel C: Average number of firms (n) and market capitalization (me)						
n	983	888	674	565	616	
me (\$10 ⁶)	1452	1373	2030	2254	3688	

Table 2: Robustness to sorting methodology & trading costs

This table evaluates the robustness of the choices made in the EBI strategy construction methodology. In each panel, the first row shows results from a quintile, value-weighted sort using NYSE break points as employed in Table 1. Each of the subsequent rows deviates in one of the three choices at a time, and the choices are specified in the first three columns. For each strategy construction methodology, the table reports average excess returns and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel A reports average returns and alphas with no adjustment for trading costs. Panel B reports net average returns and Novy-Marx and Velikov (2016) generalized alphas as prescribed by Detzel et al. (2022). T-statistics are in brackets. The sample period is 197206 to 202406.

Panel A: Gross Returns and Alphas								
Portfolios	Breaks	Weights	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}
Quintile	NYSE	VW	0.31 [3.63]	0.33 [3.87]	0.31 [3.82]	0.32 [3.86]	0.23 [2.79]	0.25 [2.92]
Quintile	NYSE	EW	0.31 [3.80]	0.35 [4.27]	0.30 [4.00]	0.26 [3.46]	0.15 [2.19]	0.12 [1.82]
Quintile	Name	VW	0.26 [2.92]	0.29 [3.15]	0.25 [2.90]	0.28 [3.11]	0.18 [2.06]	0.21 [2.31]
Quintile	Cap	VW	0.35 [4.09]	0.37 [4.20]	0.39 [4.74]	0.42 [5.01]	0.34 [4.07]	0.37 [4.36]
Decile	NYSE	VW	0.30 [2.69]	0.29 [2.60]	0.33 [3.01]	0.37 [3.25]	0.31 [2.71]	0.33 [2.92]
Panel B: Net Returns and Novy-Marx and Velikov (2016) generalized alphas								
Portfolios	Breaks	Weights	r_{net}^e	α_{CAPM}^*	α_{FF3}^*	α_{FF4}^*	α_{FF5}^*	α_{FF6}^*
Quintile	NYSE	VW	0.27 [3.21]	0.32 [3.71]	0.29 [3.59]	0.30 [3.68]	0.23 [2.82]	0.24 [2.93]
Quintile	NYSE	EW	0.11 [1.25]	0.16 [1.74]	0.09 [1.13]	0.08 [0.93]		
Quintile	Name	VW	0.23 [2.50]	0.27 [3.02]	0.24 [2.75]	0.25 [2.93]	0.18 [2.08]	0.20 [2.28]
Quintile	Cap	VW	0.32 [3.70]	0.35 [3.99]	0.36 [4.39]	0.38 [4.59]	0.33 [3.91]	0.34 [4.10]
Decile	NYSE	VW	0.26 [2.28]	0.27 [2.40]	0.30 [2.72]	0.32 [2.91]	0.27 [2.45]	0.29 [2.60]

Table 3: Conditional sort on size and EBI

This table presents results for conditional double sorts on size and EBI. In each month, stocks are first sorted into quintiles based on size using NYSE breakpoints. Then, within each size quintile, stocks are further sorted based on EBI. Finally, they are grouped into twenty-five portfolios based on the intersection of the two sorts. Panel A presents the average returns to the 25 portfolios, as well as strategies that go long stocks with high EBI and short stocks with low EBI. Panel B documents the average number of firms and the average firm size for each portfolio. The sample period is 197206 to 202406.

Panel A: portfolio average returns and time-series regression results												
EBI Quintiles						EBI Strategies						
Size quintiles		(L)	(2)	(3)	(4)	(H)	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}
	(1)	0.51 [1.86]	0.61 [2.17]	0.60 [2.24]	0.71 [2.82]	0.86 [3.33]	0.35 [3.32]	0.40 [3.77]	0.32 [3.10]	0.31 [2.97]	0.18 [1.79]	0.19 [1.81]
	(2)	0.57 [2.09]	0.65 [2.52]	0.72 [2.82]	0.73 [3.13]	0.91 [3.86]	0.34 [3.67]	0.44 [4.91]	0.35 [4.24]	0.36 [4.29]	0.18 [2.40]	0.20 [2.63]
	(3)	0.55 [2.21]	0.62 [2.59]	0.71 [3.11]	0.82 [3.79]	0.94 [4.18]	0.39 [3.91]	0.45 [4.49]	0.38 [4.01]	0.40 [4.12]	0.23 [2.53]	0.25 [2.75]
	(4)	0.57 [2.55]	0.65 [2.95]	0.71 [3.47]	0.70 [3.42]	0.94 [4.36]	0.36 [3.91]	0.40 [4.25]	0.33 [3.65]	0.39 [4.26]	0.23 [2.56]	0.28 [3.18]
	(5)	0.51 [2.54]	0.51 [2.85]	0.52 [3.05]	0.59 [3.24]	0.83 [4.11]	0.32 [3.38]	0.31 [3.25]	0.33 [3.48]	0.36 [3.71]	0.27 [2.80]	0.30 [3.06]
	Panel B: Portfolio average number of firms and market capitalization											
Size quintiles	EBI Quintiles					EBI Quintiles						
	Average n					Average market capitalization (\$10 ⁶)						
		(L)	(2)	(3)	(4)	(H)	(L)	(2)	(3)	(4)	(H)	
	(1)	425	426	426	429	431	32	33	34	37	43	
	(2)	114	114	114	114	113	61	62	63	63	64	
	(3)	79	80	80	80	79	105	109	108	109	110	
	(4)	66	66	66	66	66	222	233	229	234	234	
(5)	59	59	59	58	59	1426	1639	1590	1592	2365		

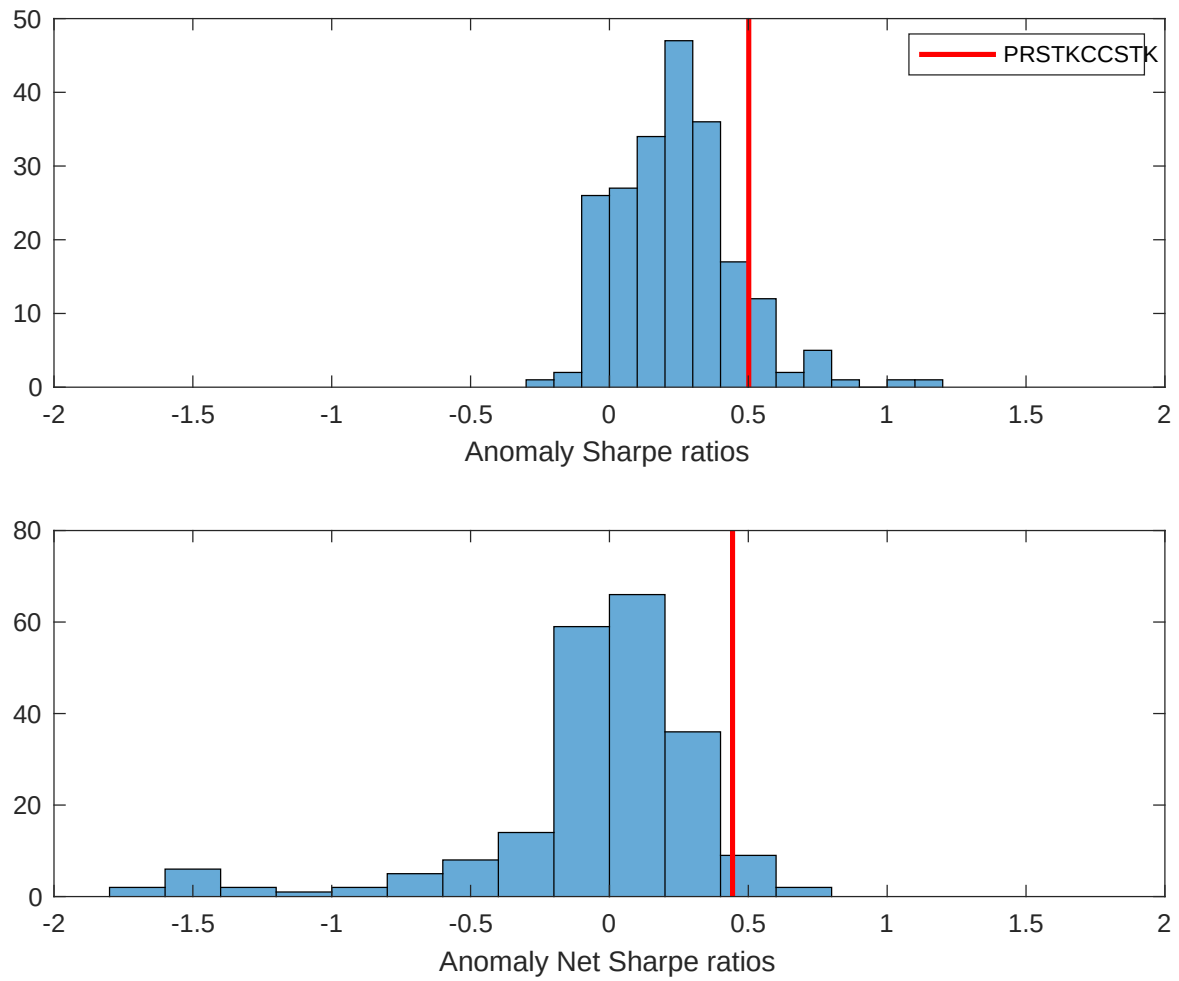


Figure 2: Distribution of Sharpe ratios.

This figure plots a histogram of Sharpe ratios for 212 anomalies, and compares the Sharpe ratio of the EBI with them (red vertical line). Panel A plots results for gross Sharpe ratios. Panel B plots results for net Sharpe ratios.

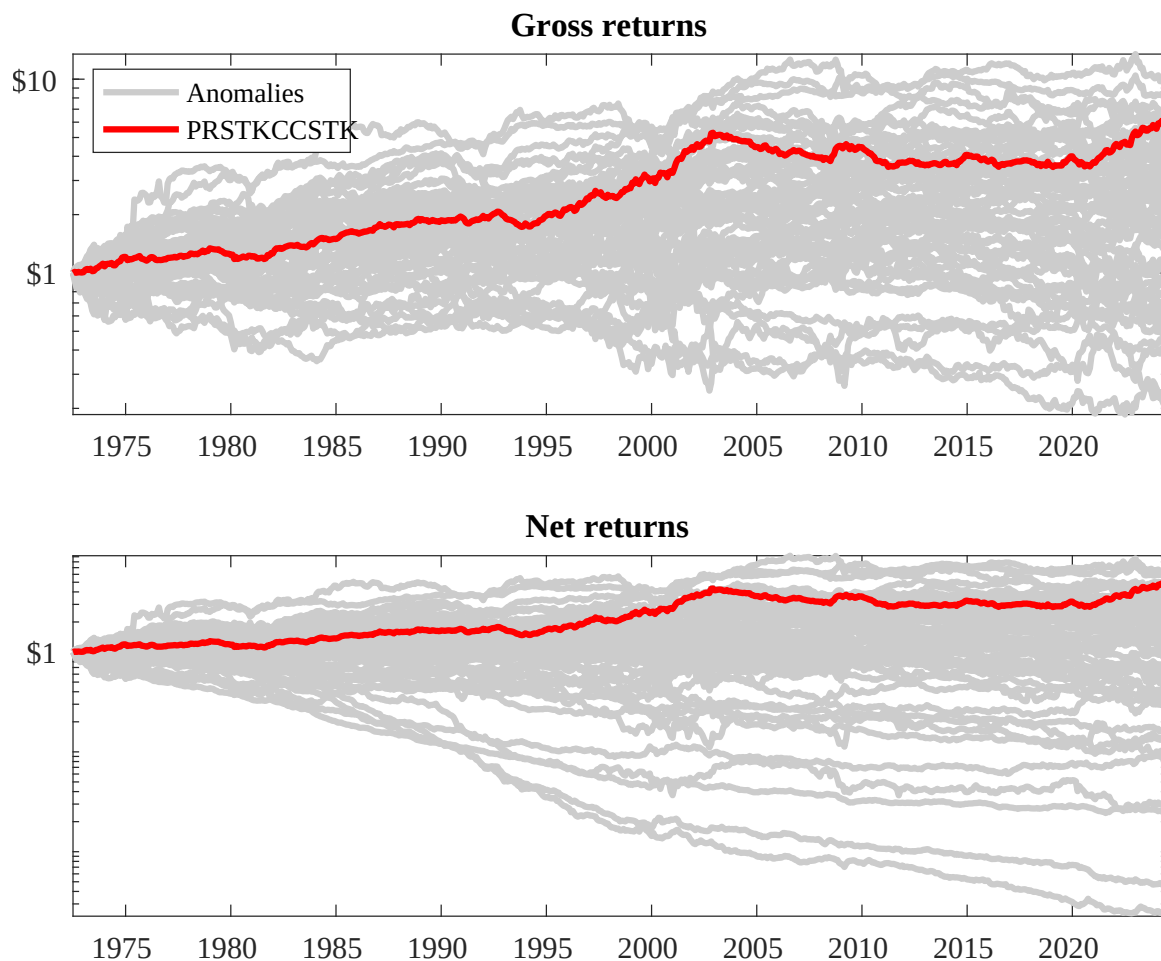
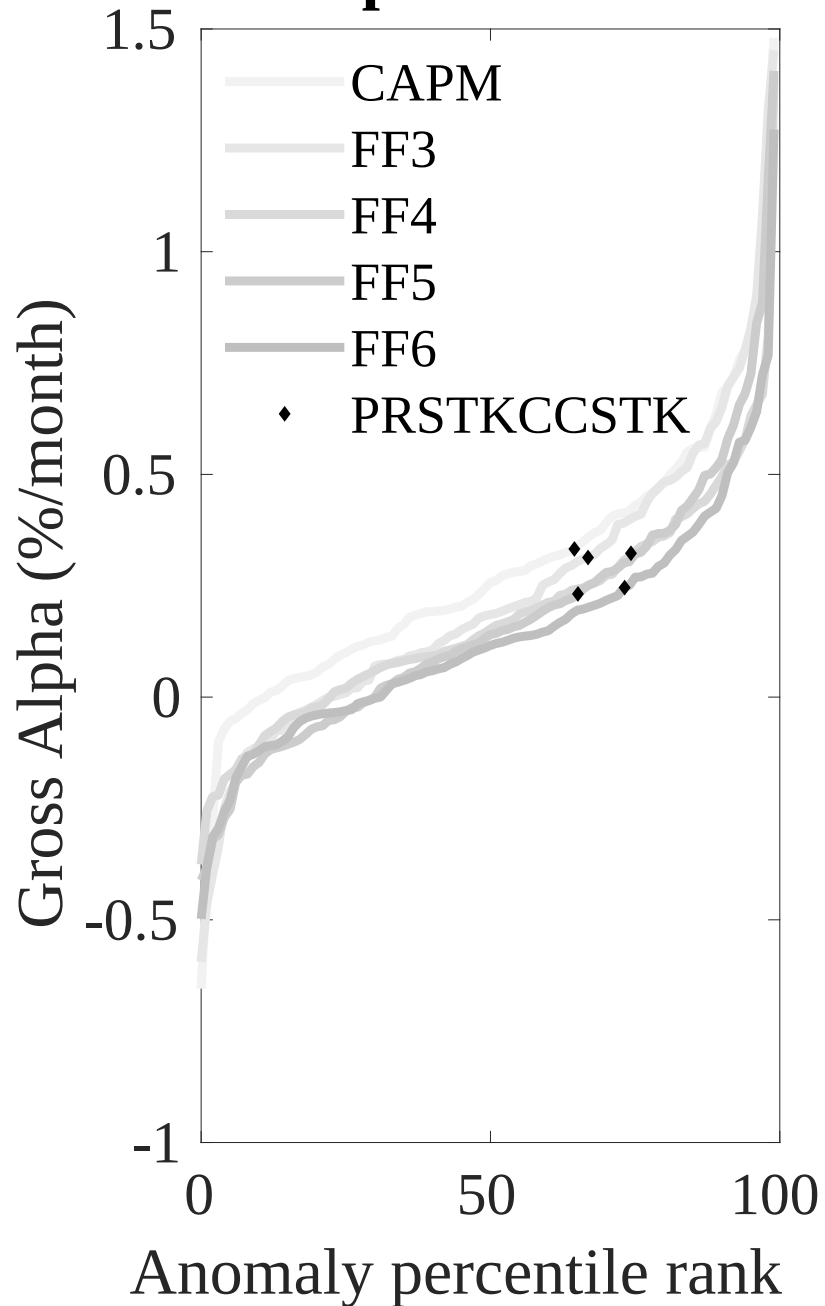


Figure 3: Dollar invested.

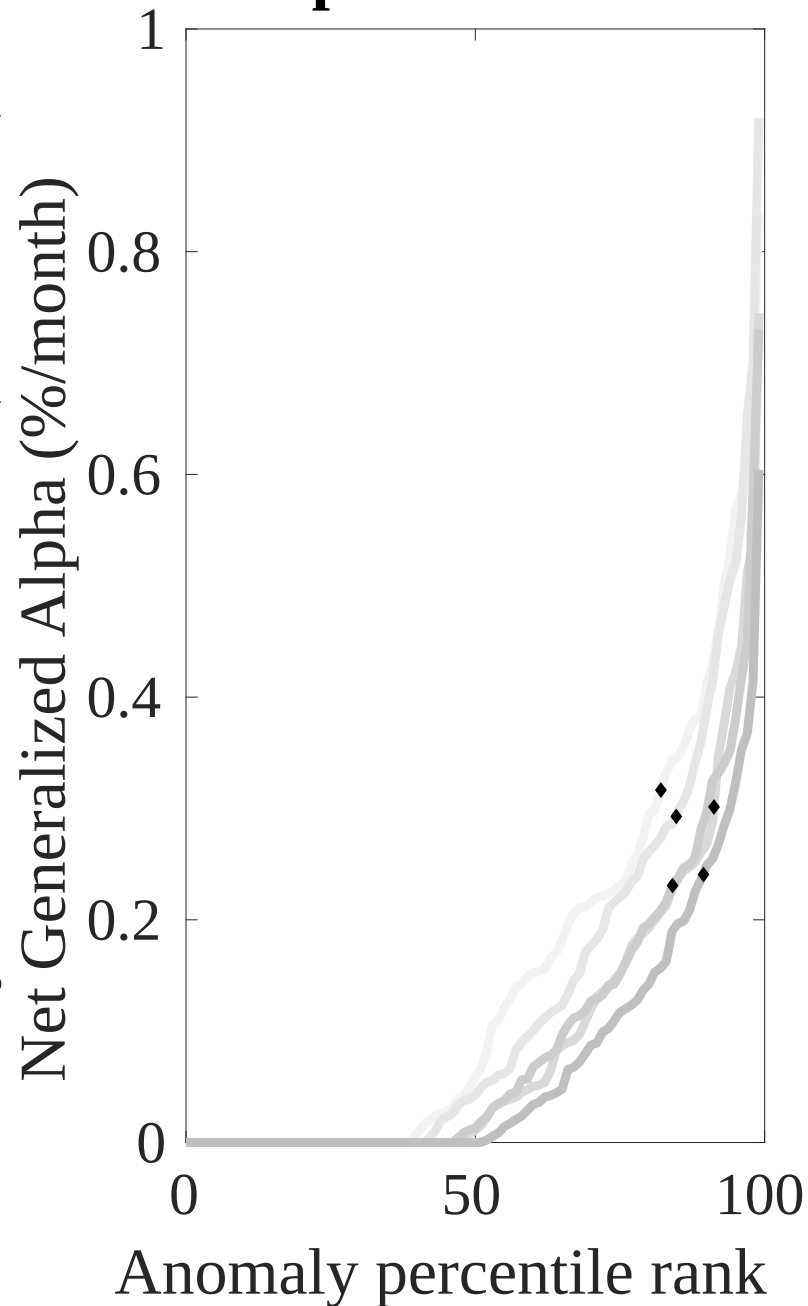
This figure plots the growth of a \$1 invested in 212 anomaly trading strategies (gray lines), and compares those with the EBI trading strategy (red line). The strategies are constructed using value-weighted quintile sorts using NYSE breakpoints. Panel A plots results for gross strategy returns. Panel B plots results for net strategy returns.

Gross Alpha Percentiles



Novy-Marx and Velikov (RFS, 2016)

Net Alpha Percentiles



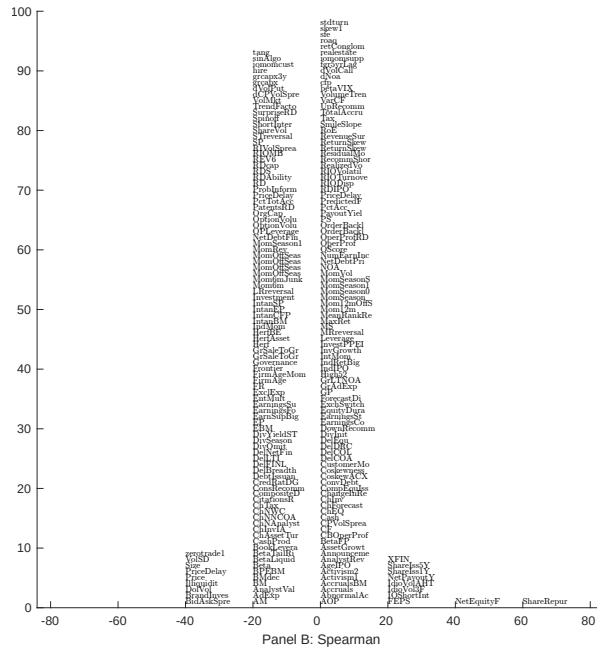
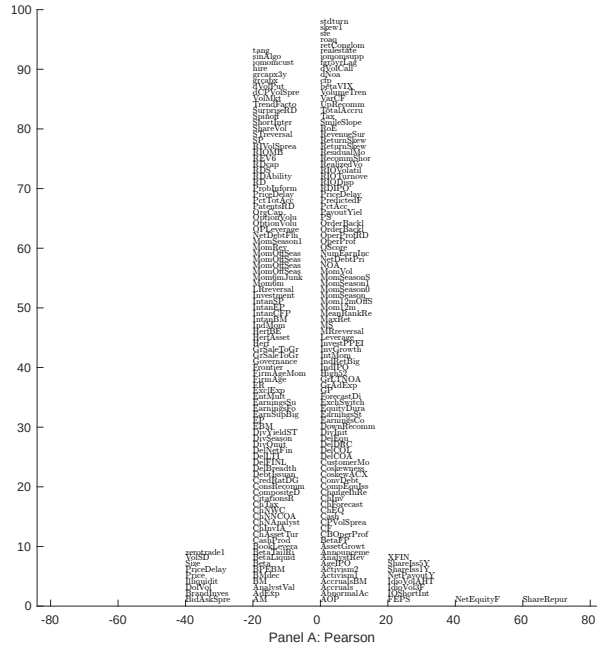


Figure 5: Distribution of correlations. This figure plots a name histogram of correlations of 210 filtered anomaly signals with EBI. The correlations are pooled. Panel A plots Pearson correlations, while Panel B plots Spearman rank correlations.

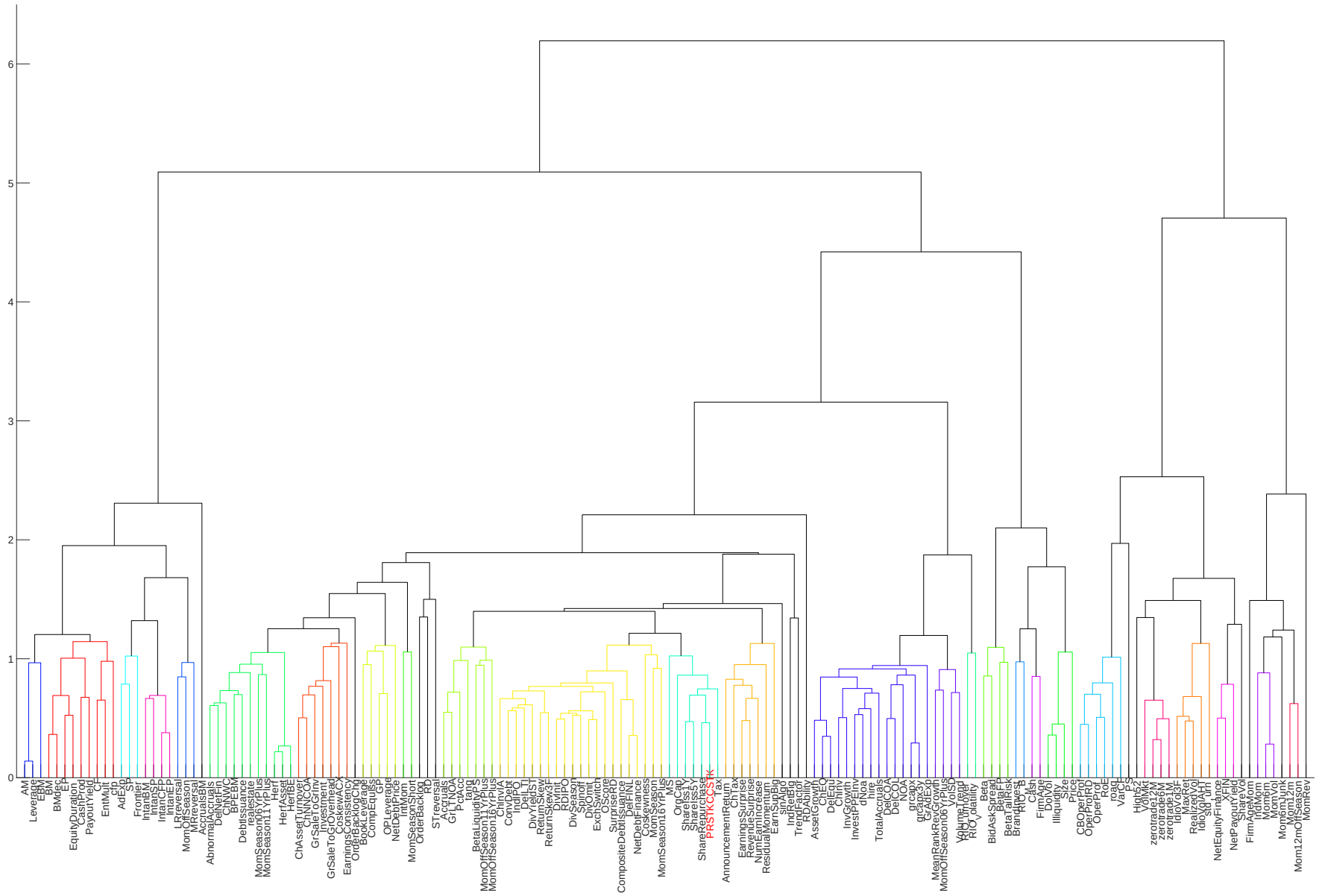


Figure 6: Agglomerative hierarchical cluster plot

This figure plots an agglomerative hierarchical cluster plot using Ward's minimum method and a maximum of 10 clusters.

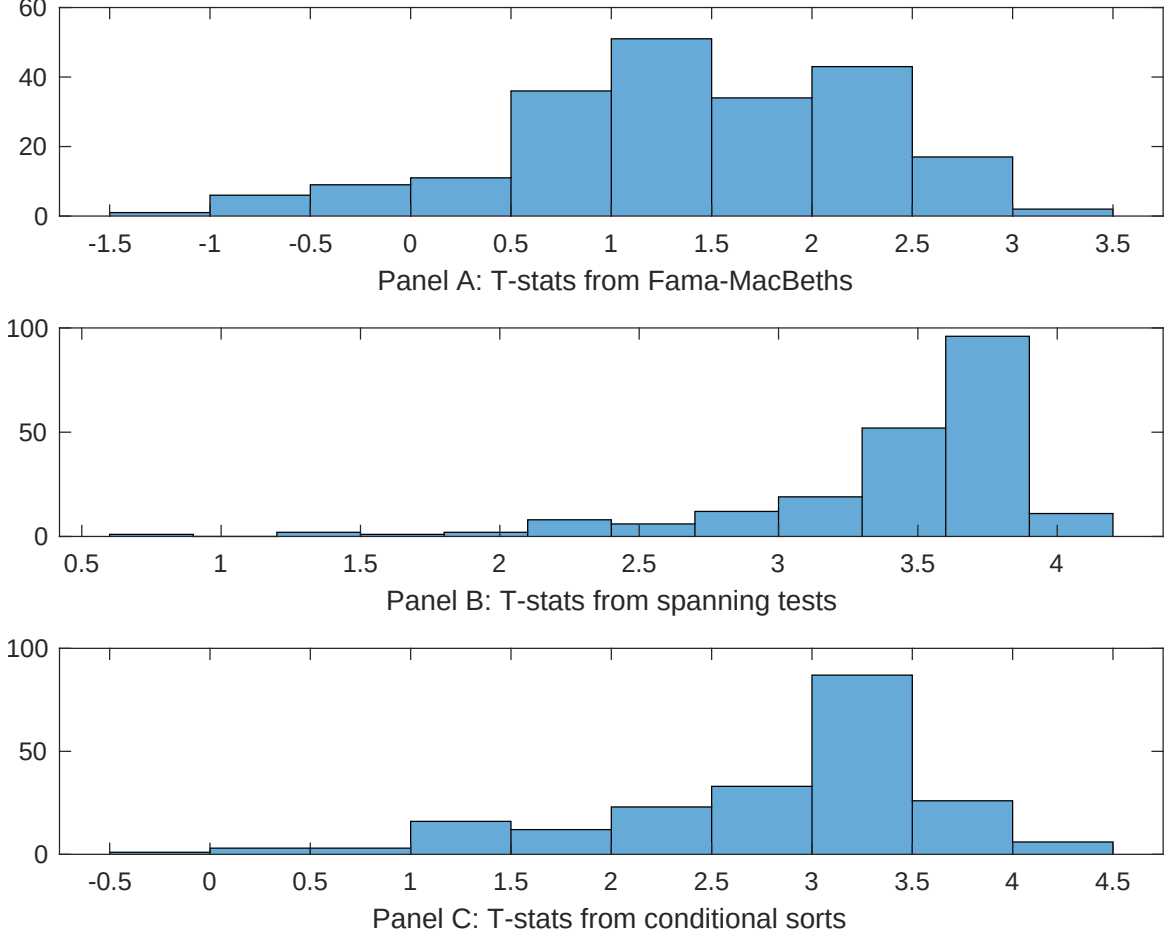


Figure 7: Distribution of t-stats on conditioning strategies

This figure plots histograms of t-statistics for predictability tests of EBI conditioning on each of the 210 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{EBI} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{EBI}EBI_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 210 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{EBI,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 210 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based on one of the 210 filtered anomaly signals at a time. Then, within each quintile, we sort stocks into quintiles based on EBI. Stocks are finally grouped into five EBI portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted EBI trading strategies conditioned on each of the 210 filtered anomalies.

Table 4: Fama-MacBeths controlling for most closely related anomalies

This table presents Fama-MacBeth results of returns on EBI. and the six most closely related anomalies. The regressions take the following form: $r_{i,t} = \alpha + \beta_{EBI}EBI_{i,t} + \sum_{k=1}^s \beta_{X_k} X_{i,t}^k + \epsilon_{i,t}$. The six most closely related anomalies, X , are Share repurchases, Net equity financing, Share issuance (1 year), Net Payout Yield, Net external financing, Share issuance (5 year). These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 197206 to 202406.

Intercept	0.10 [3.85]	0.12 [5.25]	0.13 [5.40]	0.12 [5.15]	0.13 [5.32]	0.13 [5.96]	0.13 [5.80]
EBI	-0.64 [-0.06]	0.11 [0.96]	0.13 [1.44]	1.00 [0.82]	0.68 [0.80]	0.24 [2.23]	0.73 [0.53]
Anomaly 1	0.24 [2.67]						0.54 [1.09]
Anomaly 2		0.17 [2.79]					-0.28 [-3.35]
Anomaly 3			0.17 [8.44]				0.63 [2.89]
Anomaly 4				0.28 [5.04]			0.12 [2.36]
Anomaly 5					0.19 [7.18]		0.14 [5.99]
Anomaly 6						0.32 [6.99]	0.26 [5.68]
# months	624	624	619	619	624	619	619
$\bar{R}^2(\%)$	0	1	0	1	1	0	0

Table 5: Spanning tests controlling for most closely related anomalies

This table presents spanning tests results of regressing returns to the EBI trading strategy on trading strategies exploiting the six most closely related anomalies. The regressions take the following form: $r_t^{EBI} = \alpha + \sum_{k=1}^6 \beta_{X_k} r_t^{X_k} + \sum_{j=1}^6 \beta_{f_j} r_t^{f_j} + \epsilon_t$, where X_k indicates each of the six most-closely related anomalies and f_j indicates the six factors from the [Fama and French \(2015\)](#) five-factor model augmented with the [Carhart \(1997\)](#) momentum factor. The six most closely related anomalies, X , are Share repurchases, Net equity financing, Share issuance (1 year), Net Payout Yield, Net external financing, Share issuance (5 year). These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 197206 to 202406.

Intercept	0.29 [4.23]	0.28 [3.67]	0.23 [2.99]	0.26 [3.19]	0.23 [2.87]	0.18 [2.23]	0.27 [4.05]
Anomaly 1	63.95 [18.12]						52.36 [13.38]
Anomaly 2		40.10 [11.70]					17.78 [4.03]
Anomaly 3			39.35 [10.02]				10.80 [2.45]
Anomaly 4				18.42 [5.71]			-6.10 [-1.90]
Anomaly 5					28.79 [7.08]		5.29 [1.34]
Anomaly 6						27.23 [6.15]	-3.17 [-0.74]
mkt	3.25 [2.06]	9.25 [4.97]	3.05 [1.70]	5.36 [2.72]	6.45 [3.29]	3.26 [1.73]	5.89 [3.56]
smb	-7.00 [-2.89]	-1.36 [-0.46]	-15.13 [-5.61]	-13.30 [-4.59]	-7.93 [-2.56]	-15.28 [-5.41]	-1.04 [-0.39]
hml	-6.00 [-1.97]	-2.92 [-0.86]	-0.11 [-0.03]	-6.45 [-1.68]	4.24 [1.17]	2.86 [0.79]	-4.15 [-1.30]
rmw	-3.05 [-0.93]	-7.69 [-1.91]	-7.06 [-1.68]	4.68 [1.11]	-0.31 [-0.07]	5.29 [1.30]	-15.37 [-3.92]
cma	-3.67 [-0.79]	-13.79 [-2.50]	-11.74 [-2.06]	-3.37 [-0.56]	-8.43 [-1.40]	-0.61 [-0.11]	-14.32 [-2.74]
umd	-2.42 [-1.55]	-3.04 [-1.74]	-4.06 [-2.27]	-1.15 [-0.61]	-2.54 [-1.37]	-3.78 [-2.02]	-3.46 [-2.26]
# months	624	624	620	620	624	620	620
$\bar{R}^2(\%)$	43	28	25	17	19	17	47

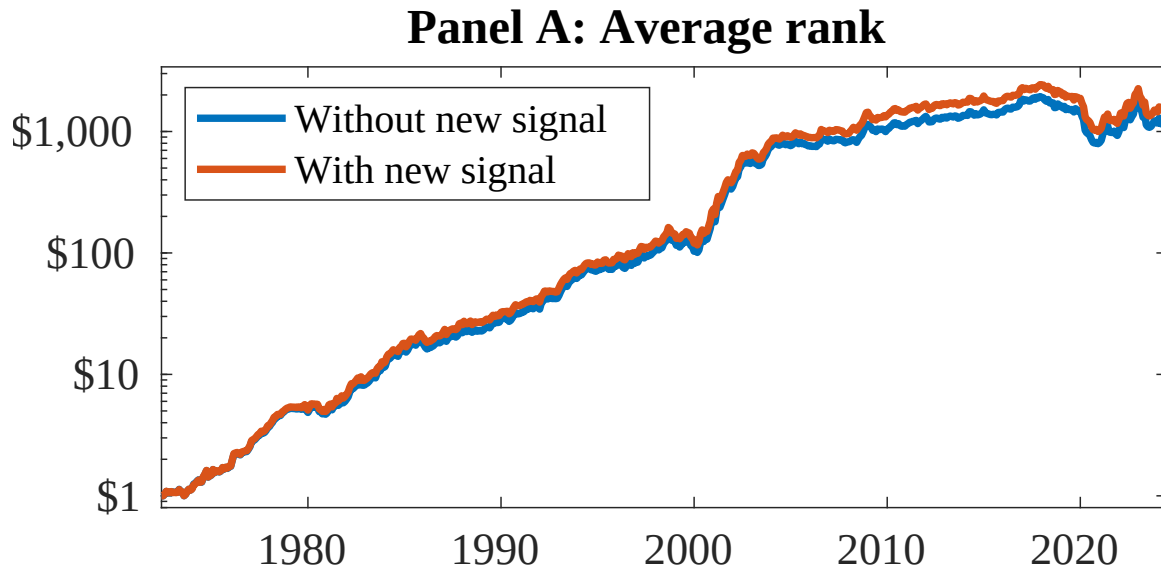


Figure 8: Combination strategy performance

This figure plots the growth of a \$1 invested in trading strategies that combine multiple anomalies following [Chen and Velikov \(2022\)](#). In all panels, the blue solid lines indicate combination trading strategies that utilize 155 anomalies. The red solid lines indicate combination trading strategies that utilize the 155 anomalies as well as EBI. Panel A shows results using "Average rank" as the combination method. See Section 7 for details on the combination methods.

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